

PORTABLE POWER STATIONS: A New Breed Of Decentralised Energy Storage Products

Portable power stations (PPS) are a new breed of products that allow significant energy storage at the user end and also allow for its use in either AC or DC form. The PPS come in various shapes and sizes, from the small ones giving about 50Wh of energy to larger stations of up to 4kWh capacity



VISHAL SAPRE is an electronics design professional working with Righill Electrics as GM (Technical). His interests include embedded control systems, IoT, power electronics, and the use of interpreted language systems, such as Python in microcontrollers

The difference between developed and emerging economies is generally expressed in terms of their per capita purchasing power parity or the per capita GDP. Another interesting data that can be considered is their per capita energy usage. Based on the freely available ample data that we have now, as shown in Fig. 1, it is clear that there has been a strong correlation between these two quantities.

Also, interestingly, not only the actual values but the changes in these two quantities are also strongly correlated, as shown in Fig. 2—particularly for the emerging economies. This suggests that the economic development of a develop-

ing country corresponds to its per capita energy usage. So, a postulation as a corollary to the above data is that, increasing the per capita energy use can result in increasing per capita GDP of the country!

Energy usage is distributed across different sources and one of the important forms of energy is electrical energy. To be able to use electrical energy, its stable availability is imperative. Over the years, it has been found that large scale power production is sustainable in the long run. However, this produced power needs to be transmitted immediately to loads scattered across the grid. In fact, power production is usually kept in sync with the on-going power demand on the grid.

But large scale energy storage is not so sustainable. Apart from operational challenges in large scale energy storage, one of the important factors is that most storage mechanisms use the direct current (DC) form, whereas the production of power and its transmission is mostly in the alternating current (AC) form.

There are inefficiencies in the conversion from AC (sources) to DC (storage) and back to AC for transmission, which compound to make this a bad choice. More so because the

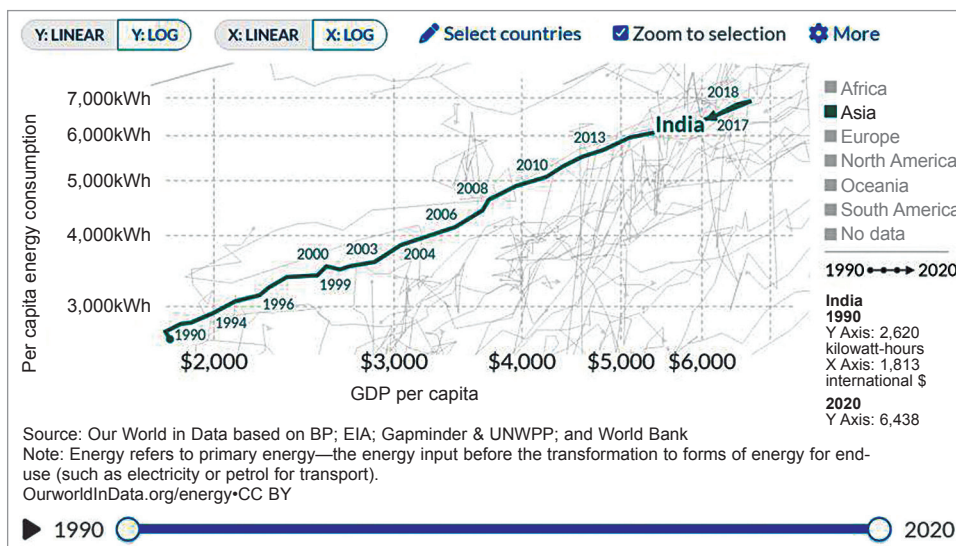


Fig. 1: Energy use per person vs GDP per capita for India (Source: <https://ourworldindata.org>)